→ Non homogeneous logistic equation

Ex: $\dot{x} = x(\mu - x) + \varepsilon, \quad \varepsilon > 0 \quad \mu \in \mathbb{R}$

Q: Modelling?

$f(x) = x\mu - x^2 + \varepsilon = -x^2 + \mu x + \varepsilon$

$\Delta = \mu^2 + 4\varepsilon > 0, \quad \forall \varepsilon > 0$

$\Rightarrow \exists \ \bar{x}_\pm = \frac{-\mu \pm \sqrt{\mu^2 + 4\varepsilon}}{-2} = \frac{\mu}{2} \pm \frac{1}{2}\sqrt{\mu^2 + 4\varepsilon}$ with $\bar{x}_+ > \frac{\mu}{2}, \ \bar{x}_- < \frac{\mu}{2}$

$f'(x) = -2x + \mu$

$f'(\bar{x}_-) = -2\bar{x}_- + \mu = +\mu + \sqrt{\mu^2+4\varepsilon} + \mu > 0 \Rightarrow$ ~~stable~~ unstable ~~for $\mu > 0$~~

$f'(\bar{x}_+) = -2\bar{x}_+ + \mu = +\mu - \sqrt{\quad} + \mu < 0 \Rightarrow$ ~~stable if $\mu < 0$~~

~~and $2\mu + \sqrt{\mu^2+4\varepsilon} < 0 \quad \sqrt{\mu^2+4\varepsilon} < -2\mu$ with $\mu < 0$~~

~~$\mu^2 + 4\varepsilon < 4\mu^2$~~

~~$3\mu^2 > 4\varepsilon \quad \mu^2 > \frac{4}{3}\varepsilon \quad \mu > \sqrt{\frac{4}{3}\varepsilon}$~~

~~→ $\exists \ \mu_c$ s.t. the stability changes~~

$f'_\mu(\bar{x}_-) = -2\left(\frac{\mu}{2} - \frac{1}{2}\sqrt{\mu^2+4\varepsilon}\right) + \mu = -\mu + \sqrt{\mu^2+4\varepsilon} + \mu > 0, \quad \forall \mu \in \mathbb{R}$

$f'_\mu(\bar{x}_+) = -2\left(\frac{\mu}{2} + \frac{1}{2}\sqrt{\mu^2+4\varepsilon}\right) + \mu = -\sqrt{\mu^2+4\varepsilon} < 0, \quad \forall \mu \in \mathbb{R}$

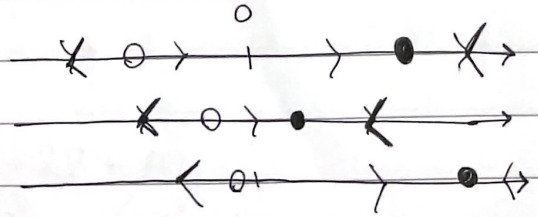
$\Rightarrow \quad \bar{x}_- \rightsquigarrow$ unstable

$\bar{x}_+ \rightsquigarrow$ stable

$\mu_c = 0 \rightsquigarrow \quad \bar{x}_\pm = \frac{\pm\sqrt{4\varepsilon}}{2} = \pm\sqrt{\varepsilon}$

$f'_{\mu_c=0}(\bar{x}_\pm) = -2\bar{x}_\pm + 0 = -2\bar{x}_\pm \Rightarrow \quad \bar{x}_+$ is stable

$\bar{x}_-$ is unstable



$\bar{x}_-$ remains unst.

$\bar{x}_+$ remains stable

Conclusion: the stability is NOT changing

Q: Modelling? Ex: $\dot{x} = x(\mu - x) - \varepsilon, \quad \varepsilon > 0, \quad \mu \in \mathbb{R}$

$f_\mu(x) = -x^2 + \mu x - \varepsilon \quad , f'_\mu(x) = -2x + \mu \quad , f''_\mu(x) = -2$

$\Delta = \mu^2 - 4\varepsilon$

* $\mu^2 - 4\varepsilon > 0 \quad |\mu| > 2\sqrt{\varepsilon} \quad (\mu < -2\sqrt{\varepsilon} \ \vee \ \mu > 2\sqrt{\varepsilon})$

$\Rightarrow \exists \ \bar{x}_\pm \in \mathbb{R} : f(\bar{x}_\pm) = 0 \quad$ with $\bar{x}_\pm = \dfrac{-\mu \pm \sqrt{\mu^2 - 4\varepsilon}}{-2}$

$$\bar{x}_\pm = \frac{\mu}{2} \pm \frac{\sqrt{\mu^2 - 4\varepsilon}}{2}$$

with $f_\mu(\bar{x}_\pm) = 0$

* $\mu^2 - 4\varepsilon = 0 \quad \Leftrightarrow \quad \mu = \pm\sqrt{4\varepsilon}$

$\Rightarrow \exists \ \bar{x} = \dfrac{\mu}{2} : f_\mu(\bar{x}) = 0$

* $\mu^2 - 4\varepsilon < 0 \quad \Leftrightarrow \quad |\mu| < 2\sqrt{\varepsilon} \quad (-2\sqrt{\varepsilon} < \mu < 2\sqrt{\varepsilon})$

$\Rightarrow \nexists \ \bar{x} \in \mathbb{R} : f_\mu(\bar{x}) = 0$

HP: $\mu > 0$ $\Rightarrow \mu_c = 2\sqrt{\varepsilon}$

Q: Do we have a tangent bif?

$f_{\mu_c}(\bar{x}) = 0 \quad , \quad f'_{\mu_c}(\bar{x}) = -2\dfrac{\mu_c}{2} + \mu_c = -\mu_c + \mu_c = 0$

$f''_{\mu_c}(\bar{x}) = -2 < 0,$

$\partial_\mu f_\mu(\bar{x})\big|_{\mu=\mu_c} = \bar{x} = \dfrac{\mu_c}{2} = \sqrt{\varepsilon} > 0$

$\Rightarrow$ bif tang.

Conclusion: with small perturbation ($\varepsilon \gtrless 0$) the qualitative analysis of the logistic equation changes!

eg. $\varepsilon > 0$ no bif. ↖

eg. $-\varepsilon < 0$ tang. bif.

Indeed, for the original case $\varepsilon = 0$, the bifurc. is of type transcritical: it does not satisfy the Hp of thm. for tangent bif.

Indeed, here in the case of transcritical bif. the quantities to take into account are (of higher order)

$$\frac{\partial f}{\partial \mu \partial x}(\mu_c, \bar{x}) \neq 0 \ ; \quad \partial_x^2 f(\mu_c, \bar{x}) \neq 0$$

EX] : Write the statement of the Prop. for transcritical bifurcation.
Proof : based on Taylor expansion.

C: Pitchfork bifurcation :

The normal form is

$$\dot{x}(t) = \mu x(t) + x^3(t) \quad , \quad \mu \in \mathbb{R}$$

$\forall \mu \in \mathbb{R}$ $\bar{x} = 0$ is always a steady state.
When μ goes trough the critical values, then $0 = \bar{x}$ changes its stability

$$f(x) = x(\mu + x^2) = 0 \iff x = 0, \forall \mu \in \mathbb{R}$$

$$x^2 + \mu = 0 \quad \nexists \, x \in \mathbb{R} \text{ if } \mu > 0$$

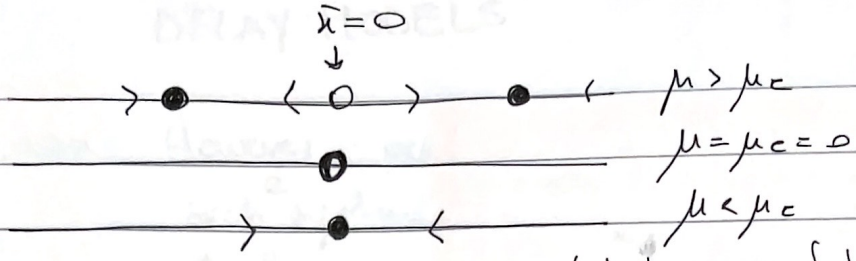
$$\bar{x}_\pm = \pm\sqrt{-\mu}, \quad \mu \leq 0.$$

$$f'(x) = \mu + 3x^2 = 0 \rightarrow \mu_c = 0, \ \bar{x} = 0$$

$$f''(x) = 6x = 0, \quad \mu_c = 0 = \bar{x} \qquad f'''(x) = 6 \neq 0$$

$$\partial_\mu f = x \quad , \quad \partial_\mu \partial_x f = 1 \neq 0$$

=> Here the main properties are $\begin{cases} \partial_\mu \partial_x f(\mu_c, \bar{x}) \neq 0 \\ f'''_{\mu_c}(\bar{x}) \neq 0 \end{cases}$



(*)

⚠ The trivial equi can be also $\neq 0$ (it depends if the system is centered at the origin or NOT). This applies to all the types of bif.

⇒ we are ready to state the following thm.

Thm (pitchfork bifurcation)

[...] Statement by ex.

Rmk To conclude: same as for the transcritical bif, the pitchfork is NOT preserved by small perturbation.

Ex: Proof